GROWTH HORMONE & ANABOLIC SIGNALING

IGF-1 LR3 — the full review

ENGINEERED LONG-ACTING IGF-1 ANALOGUE

This is the deeper look behind the short version — what the two engineered modifications actually change, what is known from the approved IGF-1 product, the hypoglycaemia risk in detail, common misunderstandings, who it is not for in full, and what your consultation covers.

What it is

IGF-1 LR3 is a laboratory-engineered version of IGF-1, the growth factor the liver produces when growth hormone signals it to. Two modifications — a substituted amino acid and an added 13-amino-acid tail — let it circulate in the bloodstream far longer and act more freely than the natural molecule.

It is sold for muscle growth and recovery. It was originally designed as a research and cell-culture reagent rather than as a medicine, and it is among the more aggressive compounds encountered on a peptide menu.

What people use it for

USED FOR	EVIDENCE	WHAT THAT MEANS
Muscle growth and hypertrophy	**Animal data only**	The mechanism is real and the animal work is clear; no human study exists.
Faster healing of tendons, muscles and injuries	**Theoretical**	Extends the mechanism beyond anything that has been tested.
Site-specific growth where it is injected	**Theoretical**	A widely repeated claim that the pharmacology does not support.

How it works

Natural IGF-1 is tightly regulated by binding proteins in the blood that limit how much of it is free to act on tissue at any given time. The engineering behind IGF-1 LR3 — the amino-acid substitution and the extended tail — reduces its binding to those regulatory proteins, which is what gives it its long half-life and its more unrestrained activity compared with the natural hormone.

Because it escapes the body's normal binding-protein restraint, IGF-1 LR3 is thought to act more potently and for longer on IGF-1 receptors throughout the body, including in muscle tissue — the basis for its use for muscle growth — but also in other tissues where unrestrained IGF-1 signaling raises safety concerns.

What the research actually looked at

The mechanism behind IGF-1 LR3's muscle-growth effect is well characterized at the biochemical level, and the animal work supporting it is clear, but no human clinical trial of IGF-1 LR3 has been conducted. Claims about faster healing of tendons, muscle, and injuries extend the mechanism beyond anything that has actually been tested.

The claim that IGF-1 LR3 produces site-specific muscle growth where it is injected — a widely repeated claim in bodybuilding circles — is not supported by how the pharmacology of a circulating protein hormone actually works; a hormone released into the bloodstream circulates systemically rather than staying localized to an injection site.

Common misunderstandings

- **“It's been tested in people.”** No human clinical trial of IGF-1 LR3 exists; the evidence base is entirely animal and laboratory work.
- **“Injecting it in one spot makes muscle grow there specifically.”** IGF-1 LR3 circulates through the bloodstream like any other protein hormone; the site-specific growth claim is not supported by how the compound's pharmacology actually works.
- **“It speeds healing of injuries.”** This extends the muscle-growth mechanism to a use that has not itself been tested, even in animals.
- **“Because it's related to a natural hormone, it's inherently safe.”** IGF-1 LR3 is engineered to escape the body's normal regulatory restraint on IGF-1, which is part of what makes it more potent — and more risky — than the natural hormone.

Who it is not for

- Anyone with an active cancer, a history of cancer, or an undiagnosed lump or symptom.
- Anyone who cannot reliably recognise and treat low blood sugar.
- Anyone taking insulin or any glucose-lowering medication, or who trains fasted or drinks alcohol.
- Anyone pregnant, trying to conceive, or breastfeeding, and any child or adolescent.

Competitive athletes: IGF-1 and its analogues, including IGF-1 LR3, fall within WADA's S2 category of prohibited peptide hormones, growth factors, and related substances; they are prohibited at all times in tested sport.

Who makes this, and under what rules

Everything dispensed here is prepared by a licensed 503A compounding pharmacy — not bought from a supplier and repackaged, and not sourced from an online seller.

That designation carries specific, enforceable obligations:

- **A prescription in your name.** Compounded from a prescription for one identified patient, after an evaluation.
- **Licensed and inspected.** A state licence, subject to inspection by the state board of pharmacy.
- **Ingredients that can be traced.** A certificate of analysis from an FDA-registered facility, back to a named source and lot.
- **Sterile preparation to a published standard.** USP standards.
- **A paper trail.** A lot number and beyond-use date on every preparation.

A compounded preparation is not an FDA-approved medicine — but it is the difference between a licensed pharmacy working to a standard and an unregulated website.

Where this stands

Never approved, and not a peptide most clinics will prescribe.

The evidence is animal and laboratory work. What is known about the risks — low blood sugar, organ enlargement, accelerating existing cancer — comes from the biology of IGF-1 itself rather than studies of this compound in people.

This is among the more potent things in the catalogue and among the least studied. Both facts matter.

Clinical detail

Engineered from native IGF-1 with a substituted amino acid and a 13-amino-acid extension (Long Arg3-IGF-1), reducing binding to regulatory binding proteins.

Originally developed as a laboratory and cell-culture research reagent, not as a therapeutic.

No published human clinical trial exists for any use.

Distinct from DES(1-3) IGF-1 and from recombinant IGF-1 — different molecules sometimes confused with one another.

Carries a hypoglycemia risk requiring the ability to recognize and treat low blood sugar.

The information on this page is general education. It is not medical advice and it is not a recommendation for any treatment.

IGF-1 LR3 is not an FDA-approved medicine for the uses described here. A compounded preparation is not an FDA-approved product and has not been reviewed by the FDA for safety, effectiveness or quality.

Descriptions of what this is used for reflect how it is commonly used and discussed. They are not claims that it is effective for any purpose.

Any prescription follows an individual evaluation by a licensed provider and is issued only where that provider judges it clinically appropriate. A prescription is never guaranteed.